

UNIVERSITY OF DHAKA

**RedGrid:
Dependency-Constrained UCB
Exploration for Autonomous
Penetration Testing**

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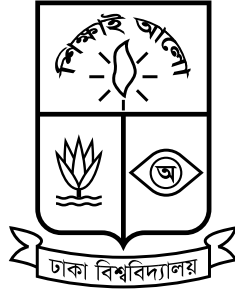
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University of Dhaka, Bangladesh

in partial fulfillment of the requirements for the degree of
Professional Masters in Information and Cyber Security (PMICS).

Declaration Form



University of Dhaka

We hereby declare that the work in this report titled “**RedGrid: Dependency-Constrained UCB Exploration for Autonomous Penetration Testing**” has been carried out by us under the supervision of Abu Ahmed Ferdaus and co-supervision of Mehedi Hasan. We further declare that where information has been derived from the literature, we have adequately cited and referenced the original sources. No part of this report was previously presented for another degree or diploma at this or any other institution.

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“To conquer fear, you must become fear.”

Ra’s al Ghul, Batman Begins

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Abstract

Abstract here ...

Acknowledgements

We would like to thank ...

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List of Algorithms

Acronym	What (it) Stands For
LAH	List Abbreviations Here

Constant Name	Symbol	=	Constant Value (with units)
Speed of Light	c	=	$2.997\,924\,58 \times 10^8 \text{ ms}^{-1}$ (exact)

symbol	name	unit
a	distance	m
P	power	W (Js ⁻¹)
ω	angular frequency	rads ⁻¹

For/Dedicated to/To my...

Chapter 1

Introduction

1.1 Background and Motivation

1.2 Problem Statement

1.3 Objectives

1.4 Scope of the Project

1.5 Expected Contributions

1.6 Challenges

1.7 Organization

The thesis is organized in a systematic way. The organization can be seen as:

Chapter 2

Literature Review and Related Work

2.1 Domain Overview

2.2 Existing Systems

2.3 Comparative Analysis

2.4 Research Gap

2.5 Summary

Chapter 3

Proposed Methodologies

3.1 System Overview

3.2 Requirement Analysis

3.3 Proposed Architecture

3.4 System Workflow

3.5 Tools and Technologies

Chapter 4

Execution Plan

4.1 Work Breakdown Structure

4.2 Project Timeline

4.3 Milestones

4.4 Resource Requirements

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Chapter 5

Experimental Results

5.1 Evaluation Plan

Chapter 6

Conclusions

6.1 Summary

6.2 Expected Deliverables

6.3 Future Works

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